Order and Degree of a Differential Equation

Once you have identified an equation as a Differential Equation (DE), the next critical step is to classify it by its **Order** and **Degree**. These two properties determine the complexity of the equation and dictate which mathematical methods you will need to use to solve it.

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1. Concept Overview

1.1 The Order of a Differential Equation

> **Definition:** The **Order** of a differential equation is the order of the highest differential coefficient (derivative) present in the equation.

- If the highest derivative is $\frac{dy}{dx}$ (or y'), it is a **1st-order** DE.
- If the highest derivative is $\frac{d^2y}{dx^2}$ (or y''), it is a **2nd-order** DE.
- If the highest derivative is $\frac{d^ny}{dx^n}$ (or $y^{(n)}$), it is an **n-th order** DE.

1.2 The Degree of a Differential Equation

> **Definition:** The **Degree** of a differential equation is the highest power (exponent) of the highest-order derivative present in the equation.

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2. Important Working Rule

To accurately find the degree of a differential equation, the equation **must be made free from radicals and fractions** as far as the derivatives are concerned.

If a derivative is inside a square root or raised to a fractional power (e.g., 3/2), you must use algebraic manipulation (like squaring or cubing both sides) to clear those fractional powers before determining the degree.

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3. Solved Questions

Here are step-by-step examples from the lecture notes demonstrating how to find the order and degree.

Question 1

Determine the order and degree of the following Differential Equation:

$$\cos x \frac{d^2y}{dx^2} + \sin x \left(\frac{dy}{dx}\right)^3 + 8y = \tan x$$

Solution: 1. **Find the Order:** Look for the highest derivative in the equation. * We have a first derivative: $\frac{dy}{dx}$ * We have a second derivative: $\frac{d^2y}{dx^2}$ * The highest derivative is the second derivative. Therefore, the **Order is 2**. 2. **Find the Degree:** Look at the power of the *highest* derivative. * The highest derivative is $\frac{d^2y}{dx^2}$. * It is not raised to any explicitly written power, which means its power is 1. * *(Note: Do not be tricked by the $\left(\frac{dy}{dx}\right)^3$ term. Even though it has a higher power of 3, it is not the highest-order derivative).* 3. **Final Answer:** * **Order:** 2 * **Degree:** 1

Question 2

Determine the order and degree of the following Differential Equation:

$$\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}=K\frac{d^2y}{dx^2}$$

Solution: 1. **Apply the Working Rule (Clear Radicals):** We notice that the derivatives on the left side are raised to a fractional power (3/2). We cannot determine the degree until this fraction is removed. 2. **Algebraic Manipulation:** To eliminate the denominator of the fraction (/2), we square both sides of the equation.

$$\left(\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}\right)^2=\left(K\frac{d^2y}{dx^2}\right)^2$$

$$\left[1+\left(\frac{dy}{dx}\right)^2\right]^3=K^2\left(\frac{d^2y}{dx^2}\right)^2$$

3. **Find the Order:** Now look at the manipulated equation. The highest derivative present is $\frac{d^2y}{dx^2}$. * Therefore, the **Order is 2**. 4. **Find the Degree:** Look at the power of that highest derivative. * The term $\left(\frac{d^2y}{dx^2}\right)$ is raised to the power of 2. * Therefore, the **Degree is 2**. 5. **Final Answer:** * **Order:** 2 * **Degree:** 2

Next Topic: Now that you know how to classify DEs, proceed to Topic 1.3: Formation of Differential Equations to learn how these equations are originally constructed.